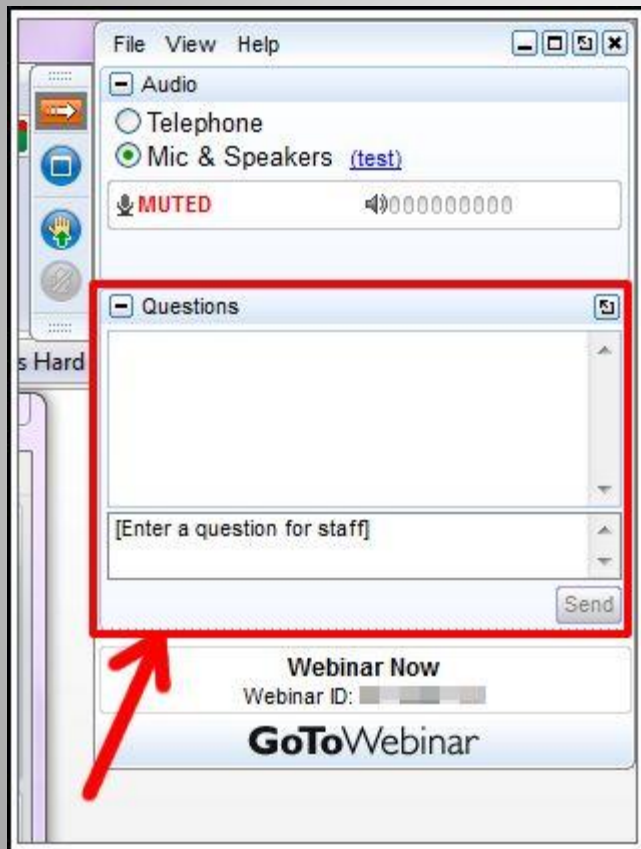




Building Fatigue Resistance

Tim Cusick | WKO4 Training Strategy

Questions



- Everyone on mute
- Type questions; we'll get to as we go or at end
- We'll email replies if too many questions

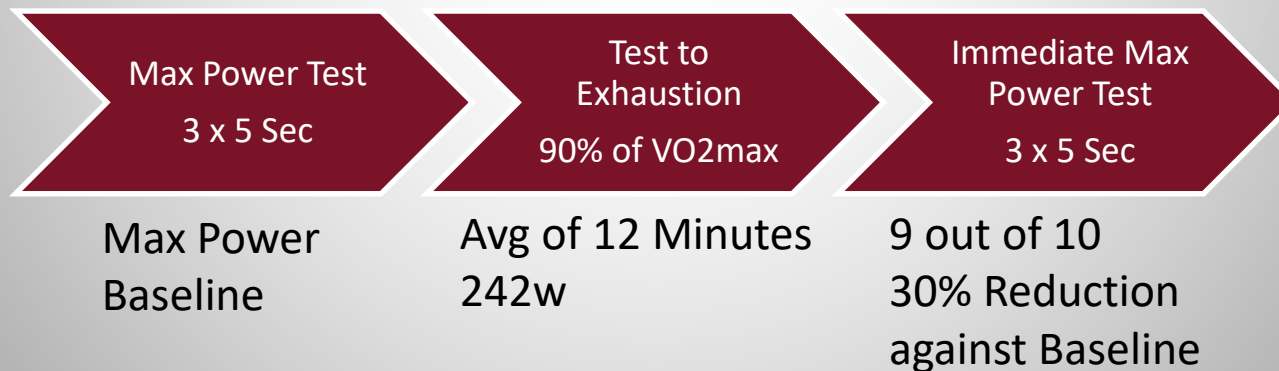
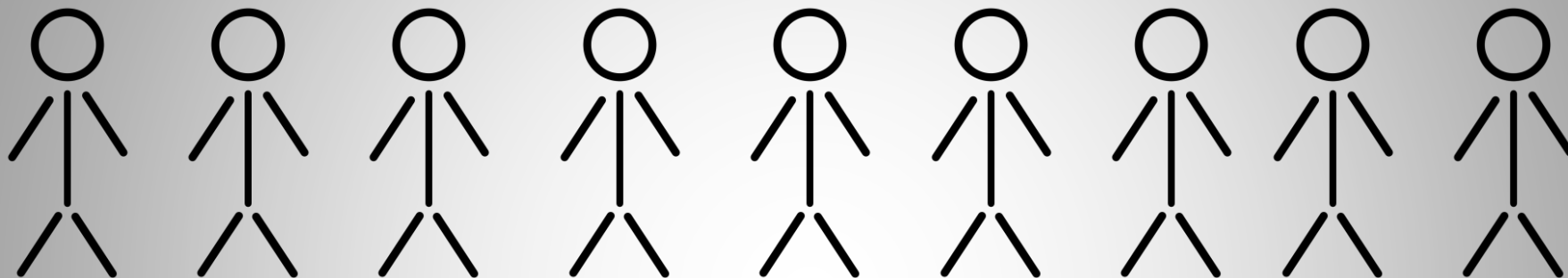


“Yes, everyone has one, and this is mine.”

...first, a valuable lesson about
Fatigue Resistance

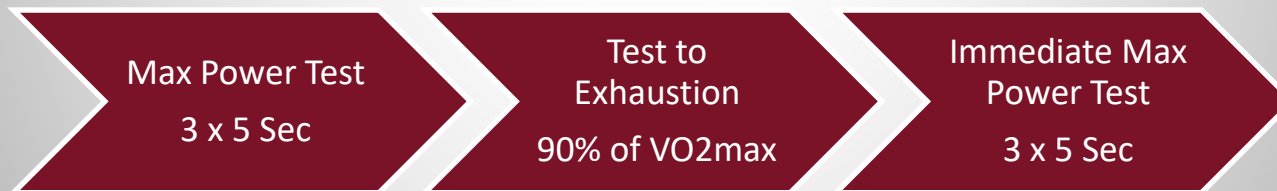
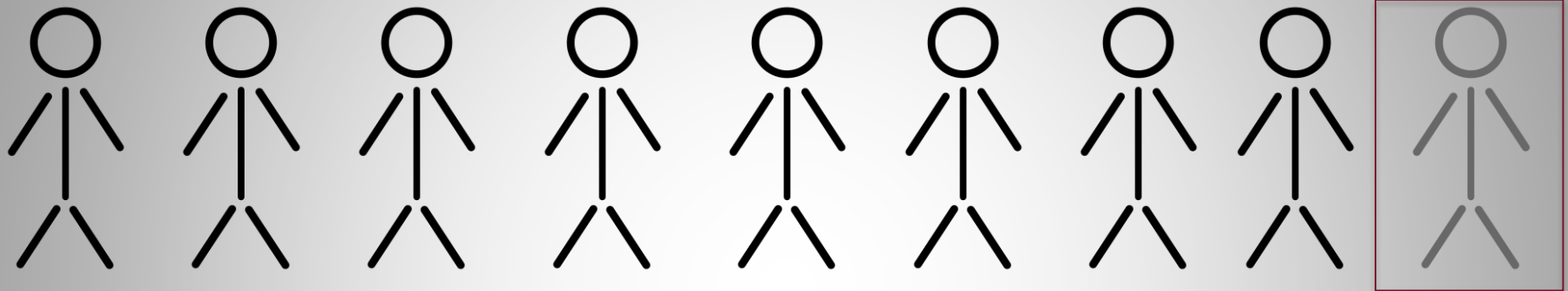
A lesson about Fatigue Resistance

10 People



A lesson about Fatigue Resistance

10 People



Max Power
Baseline

Avg of 12 Minutes
242w

9 out of 10
30% Reduction
against Baseline

That Guy
90% +
reduction
in power

THAT
GUY WAS
THE
BEST!

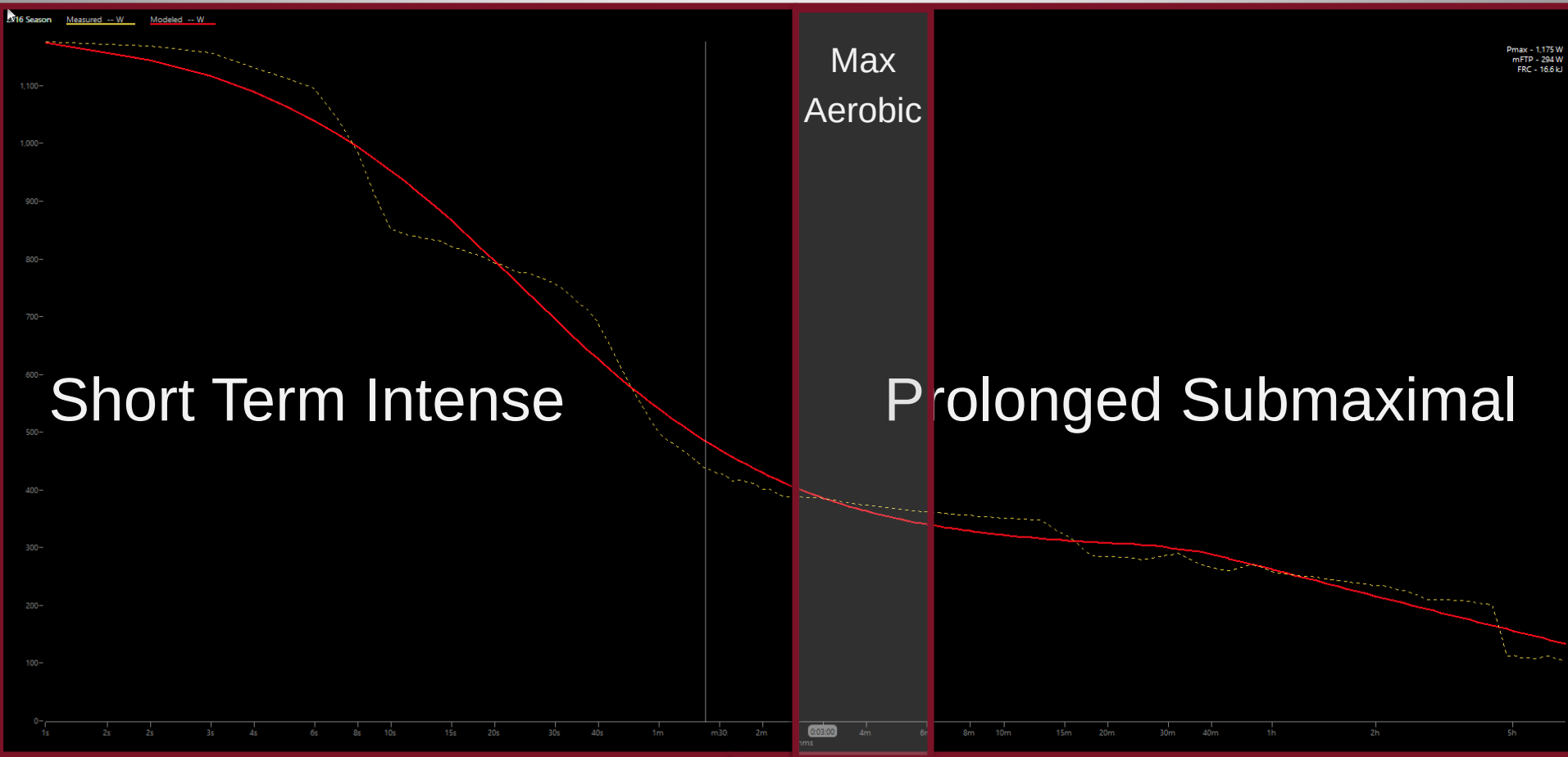
...question the
20 minute revolution

Fatigue

1. An exercise-induced progressive decline in maximal force production
2. The point when you can no longer maintain a given output of force

Cycling Fatigue

The two dominant sport conditioning scenarios leading to fatigue are short-term intense exercise and extended, submaximal training events.



2-3 Min

Where does fatigue occur?

Central or neural fatigue (CNS)

- Proximal to the motor unit
- Limitation of the nerves ability to generate sustained signal

Peripheral or metabolic fatigue

- Residing within the motor unit
- Reduced ability of the muscle fiber to contract
- The “motor unit”

What Causes Fatigue?

Two possible ways to look at it:

- The Accumulation hypothesis
 - The Depletion hypothesis
-
- *The origin of fatigue is exercise-dependent and may be due to either accumulation, depletion, or both.*

Accumulation Hypothesis

- There is a **buildup of metabolic by-products** in the muscle fiber
 - Lactic acid (lactate)
 - Hydrogen ions (H^+)
 - Ammonia
 - Inorganic phosphate
- **Lactate is the primary marker** associated with the accumulation hypothesis
- If you exercise at a **high enough intensity**, H^+ **accumulation** interferes with force production

Depletion Hypothesis

Aspects to the **depletion hypothesis**:

- Neural depletion
 - Depletion of acetylcholine
 - a chemical released by nerve cells to send signals to other cells
- Depletion of **energy substrates**
 - ATP depletion
 - Phosphagen depletion
 - Glycogen depletion
 - *Basically, you run out of fuel*

What Type of Fatigue

Short Term Intense

- Creatine phosphate depletion
- Decrease in ATP
- Lactate accumulation
- Proton accumulation
- Decrease in neural muscular activity

Prolonged Submaximal

- Decreased blood glucose
- Decrease muscle glycogen
- Dehydration (?)
- Hyperthermia
- Lactate accumulation

Fatigue Resistance

1. Resist or slow the rate of exercise-induced progressive decline in maximal force production
2. Ability to delay (extend) the point when you can no longer maintain a given output of force

So what is the best way to
use this knowledge?

...increase the odds
of success

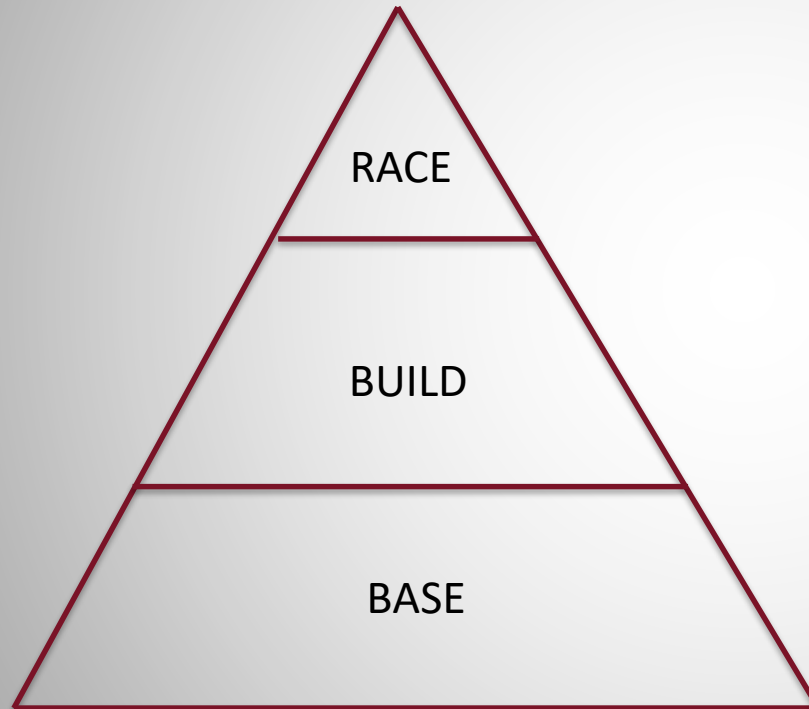
Improving the odds of success: fatigue resistance

Functional threshold power is the single greatest factor related to performance success in endurance sports but lack of fatigue resistance is the greatest factor in performance failure

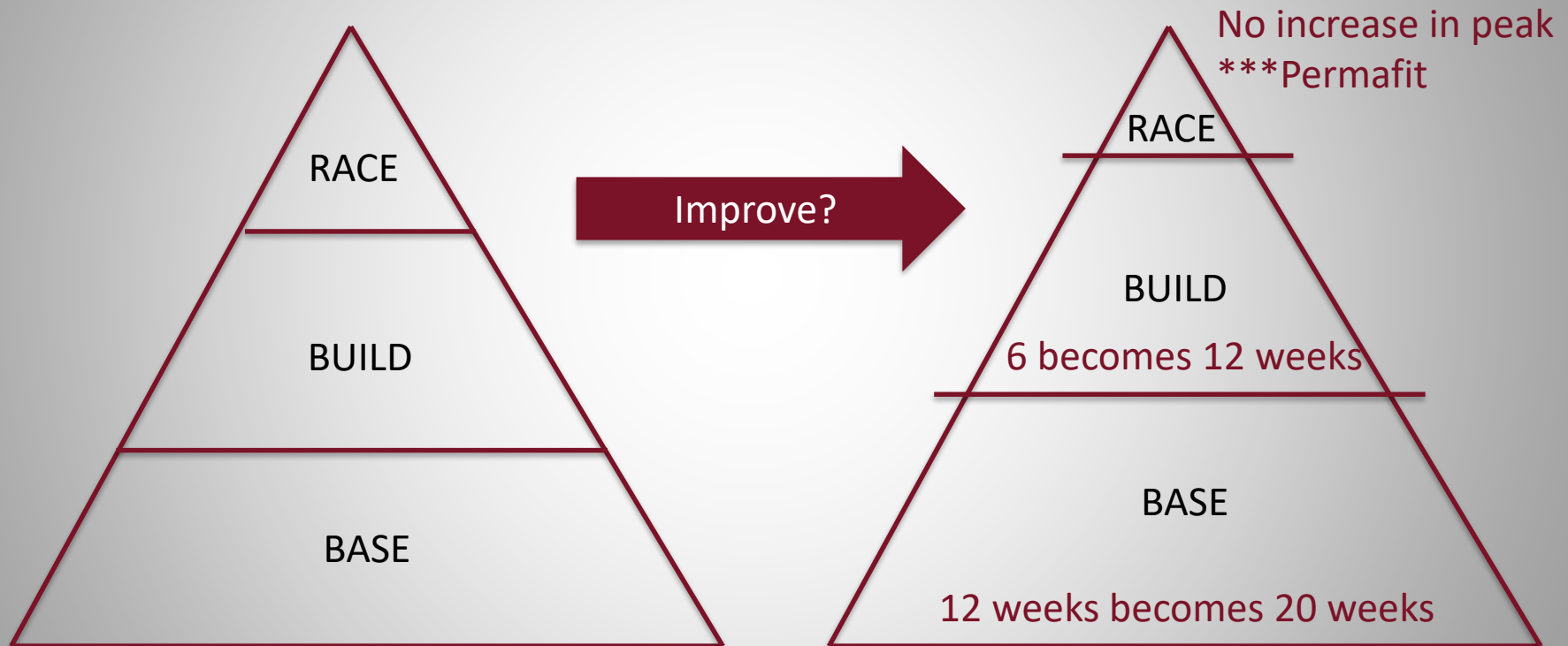
Fatigue Resistance Strategy

- Add to **annual training** plan thinking
- Utilize understanding to build **better training tactics and workouts**
- **Motivate athletes** in a carrot and a stick fashion that is not just “more power”
 - A reason, an answer to “why”
- Improving the ability to “stack” **periodized training**, the more you train, the more you can train

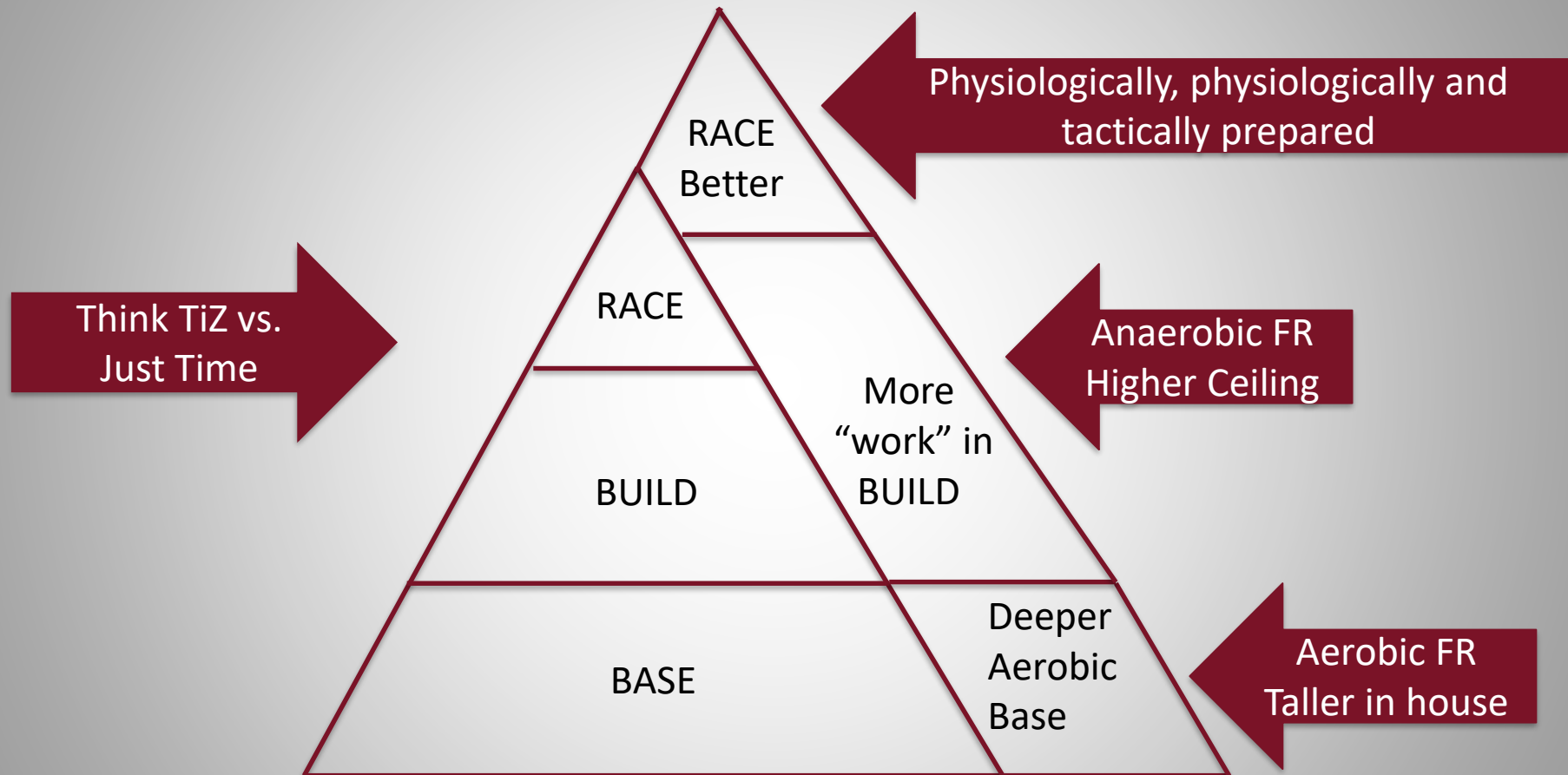
Stack Periodization Annual Training



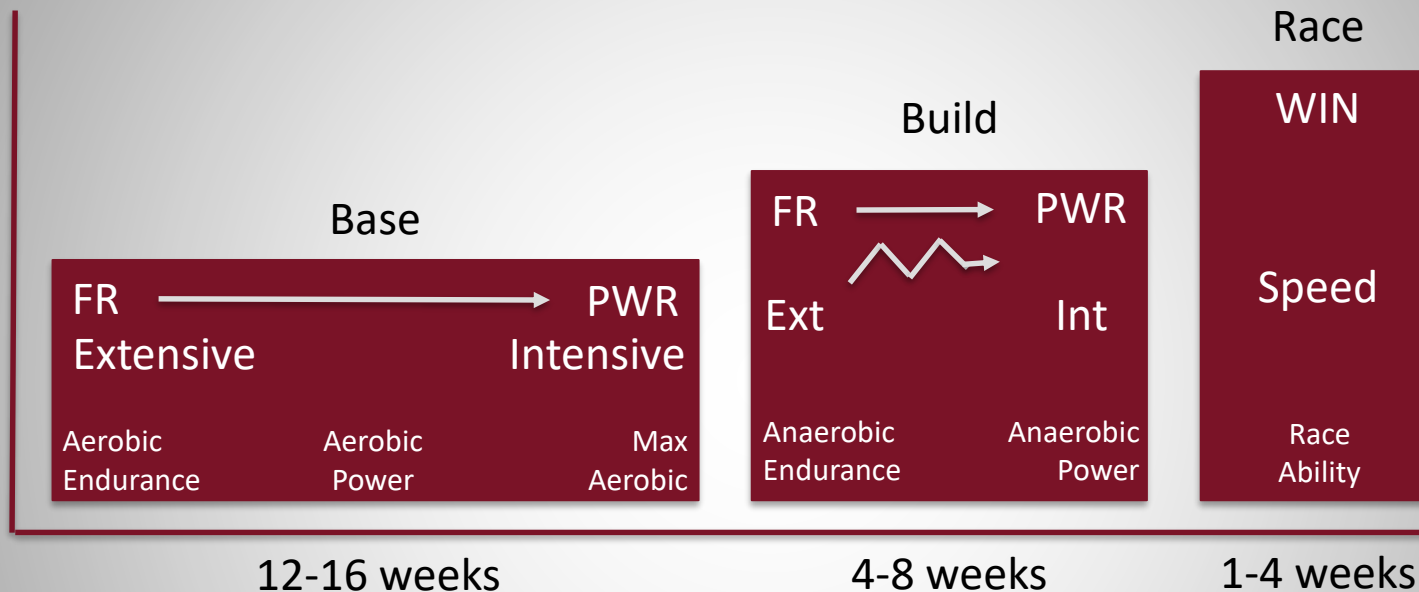
Stack Periodization Annual Training



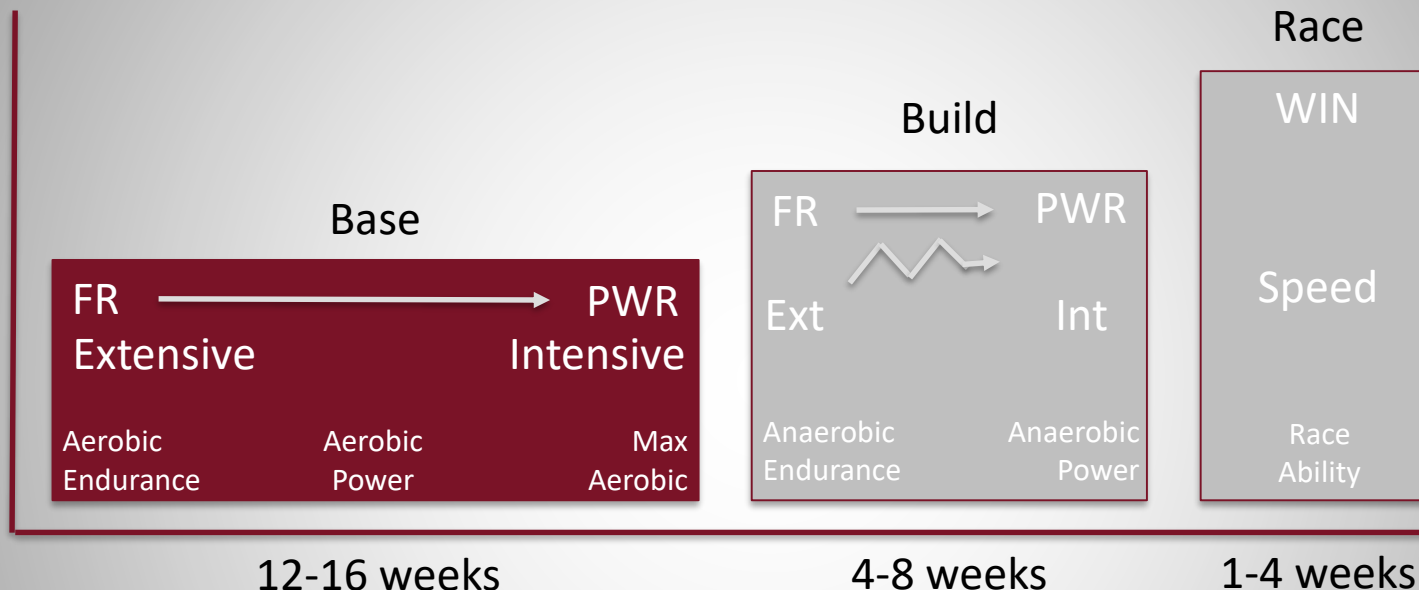
Stack Periodization Annual Training



Periodization Priority for Fatigue Resistance



Periodization of Priority for Aerobic Fatigue Resistance



The “Building Blocks of More”

Maximize Duration

Maximize “Disciplined TiZ”

Progression principle driven by TiZ initially, moves towards power

Progression Principle:

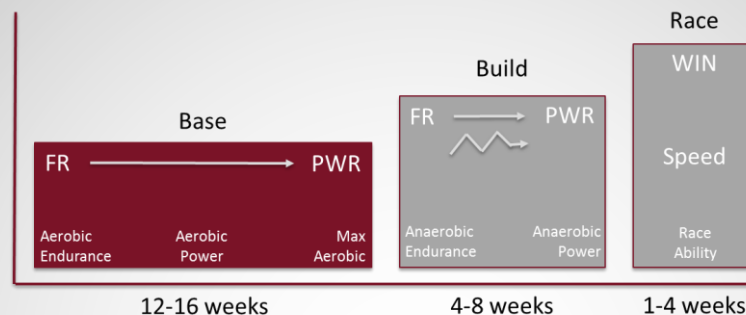
Progress TiZ as driver

Power increase secondary and follows

Base Targets

CLASSIC LEVELS	TRAINING TARGET	COGGAN iLEVELS
Recovery 1		1 Recovery
Endurance 2	Aerobic endurance (extensive)	2 Endurance
Tempo 3		3 Tempo
Lactate threshold (FTP) 4		4a Sweet spot
VO2Max 5	Aerobic power (intensive)	4 Threshold (FTP)
	Max aerobic power	5 FRC/FTP
Anaerobic capacity 6	Anaerobic endurance (extensive)	6 FRC
	Anaerobic power (intensive)	7a Pmax/FRC
	Maximal power	7 Pmax
(Neuromuscular power 7)		

Base



	Aerobic Endurance	Aerobic Power	Max Aerobic
Core	Duration is king	TiZ: Tempo / SST FTP	Max Aerobic Intervals (VO2max)
Focus	Mid Endurance Zone	Progress time not power (power will follow)	
Frequency	Always 😊	3-5 x week	Event/Rider specific this phase, 1x every 7-14 workouts
Secondary Focus	Limiter	Limiter	Event Demand / Limiter

Base: Fatigue Resistance Tactics

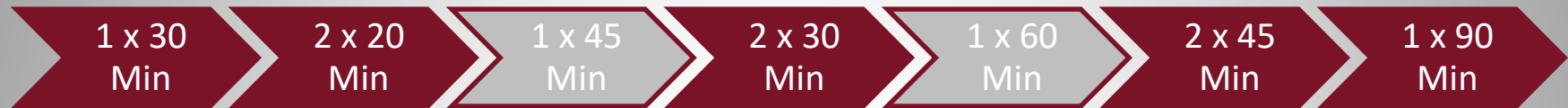
Progression principle rules

- Aerobic / Muscular Endurance = Fatigue resistance
- Progress through training levels in general
- Progress time AT training level / target as priority

Power levels / targeting important here

- Mid Level 2
- True Tempo
- True SST
- THEN FTP work

Example: Base Tempo Progression

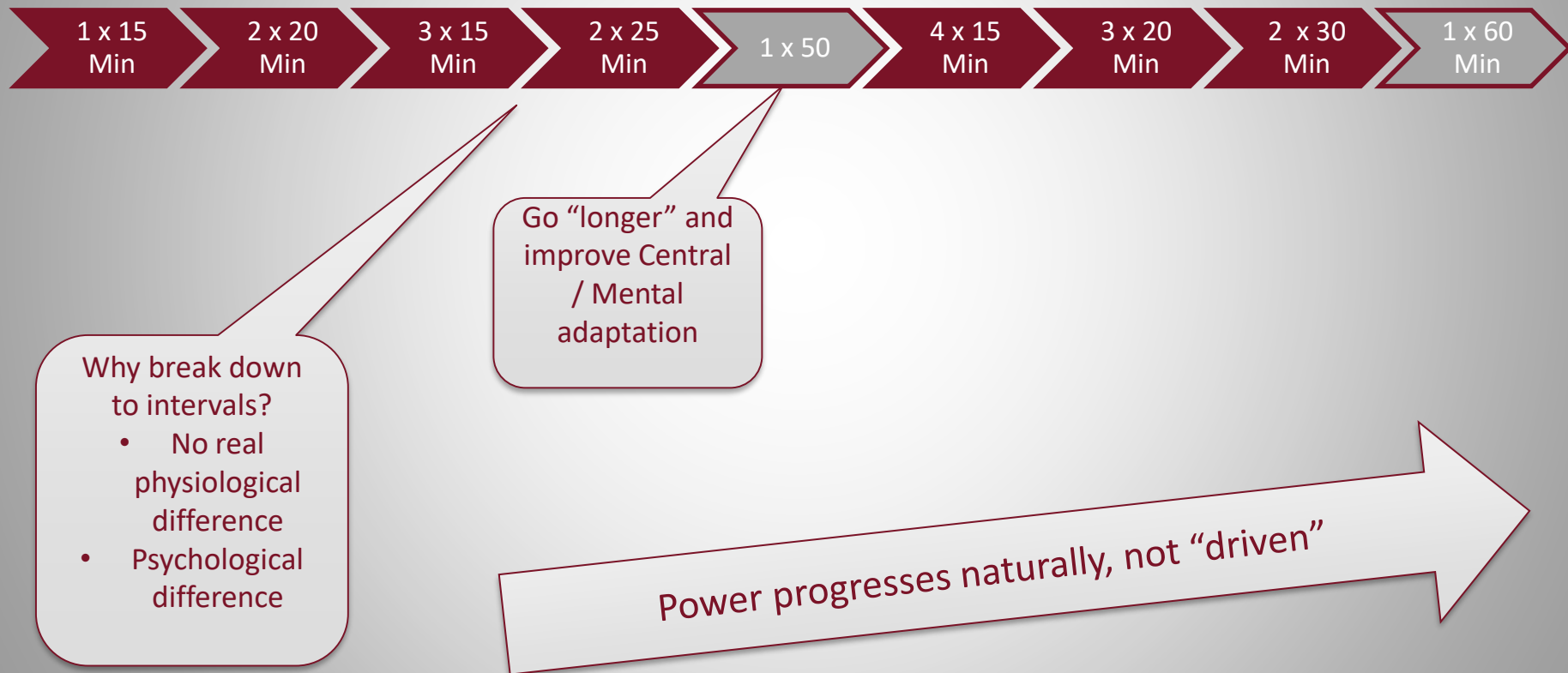


Power progresses naturally, not “driven”

Why break down to intervals?

- No real physiological difference
- Psychological difference
- Each step prepares for next

Example: SST Progression



Example: FTP Progression



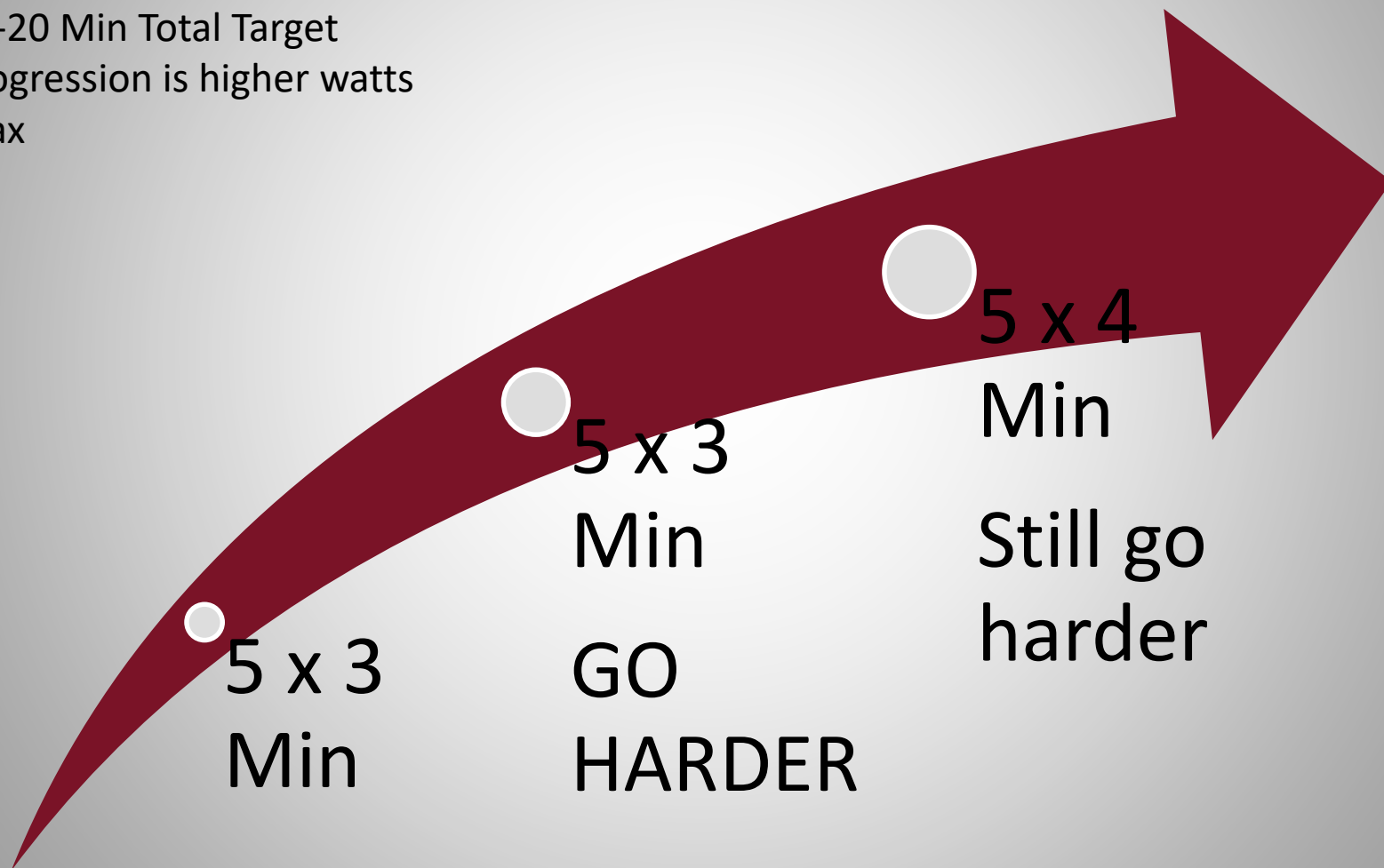
Example: Max Aerobic

Classic VO2max

15-20 Min Total Target

Progression is higher watts

Max



Total Intensive Aerobic by Workouts Goals

	Rec / Century	Cat 3–5	Cat 1-2	Pro
Tempo Total Duration	Up to 60 Min	60-90 Min	90 – 180 Min	180 Min++
SST Total Duration	Up to 45 Min	45-60 Min	60-80 Min	60 min +++
FTP Total Duration	Up to 40 Min	40-60 Min	60 Min +	60 Min ++
Max Aerobic	10+ Min but target 15 Min	15 Min (max) 15-20 Min	15 Min (max) 15-25 Min	15 Min (max) 15-25 Min

Guidelines by my opinion

Base: Fatigue Resistance Workout

EPD SST INTERVALS

WU:

10-15 Min Warm Up, progressing to Zone 2

TARGET: (PW: Z2 / HR: Zone 2 / PE: 4-5)

CADENCE: Self Selected with 2 x 1 minute Fast Pedals to wake up legs

MS1:

2 x 16 min EPD SST Intervals

Start each interval with 3 x 12 sec max sprint with 24 sec recovery between sprints. IMMEDIATELY following third sprint (DO NOT REST) complete a 15 Minute Interval in the FTP Zone

TARGET: FTP (PW Z4/HR Z4/RPE 6-7)

REST: 10 min between intervals, easy riding

CADENCE: Self-selected but do not let RPM drop as you fatigue!

CD: 5-15 Minutes of easy pedaling in L1 (Power Z1, HR Z1, RPE <2)

SST Crisscross Intervals

WU:

10-15 Min Warm Up, progressing to Zone 2

TARGET: (PW: Z2 / HR: Zone 2 / PE: 4-5)

CADENCE: Self Selected with 2 x 1 minute Fast Pedals to wake up legs

MS1:

2 x 20 min Crisscross Sweet Spot Intervals.

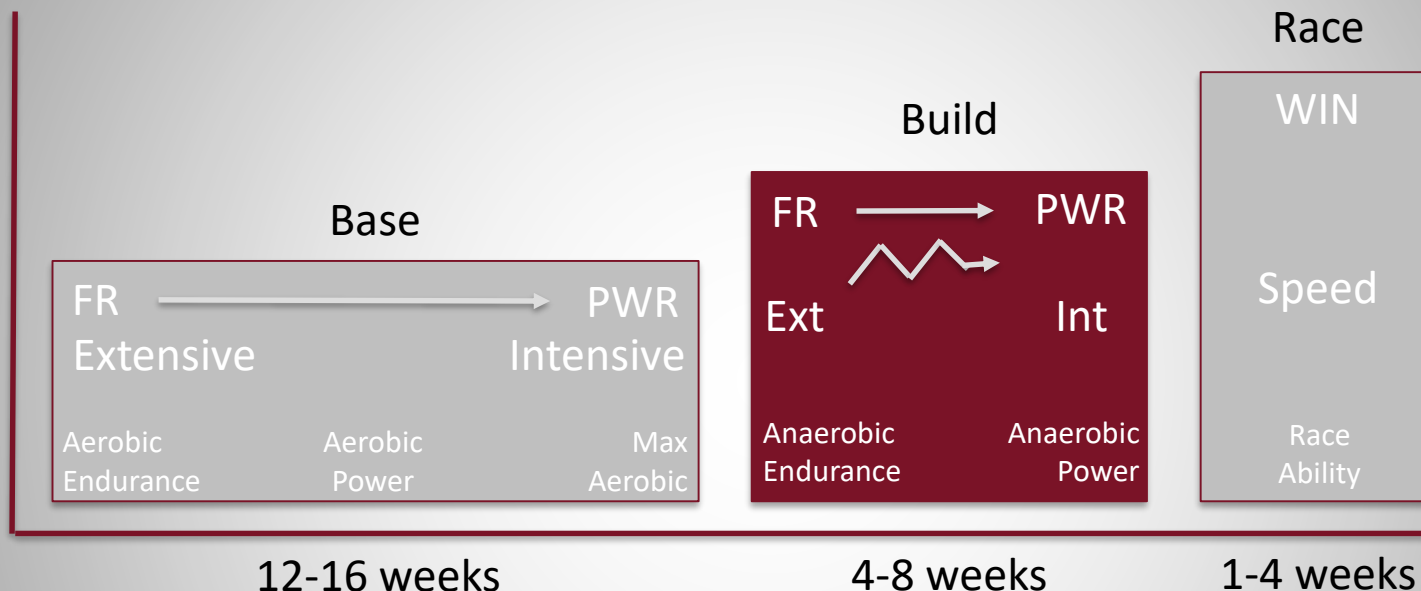
A “crisscross” interval is an interval where you vary your efforts to really teach your body to clear lactate. Every 2 minutes during the SSI interval, increase effort to 110% of FTP for 30 seconds then return to SSI power level. Do not drop off (below 88% of FTP) or recover till interval is complete.

TARGET: 88-93% of FTP / SSI (PW: Level 3.5 / HR Zone 3.5 / PE 6-7)

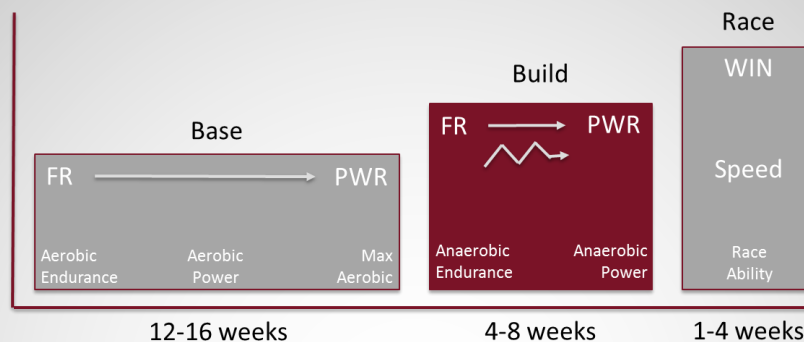
CADENCE: Self selected

CD: 5 - 15 Minutes

Periodization of Priority for Anaerobic Fatigue Resistance



Build



	Anaerobic Endurance	Anaerobic Power	Max Aerobic
Core	Focus on max (ALERT)	Focus on max	Max Aerobic Intervals (VO2max)
Focus	FRC	Pmax/FRC	
Frequency	1-3 per week	1 – 2 per week	Event/Rider specific this phase, 1x every 7-14 workouts
Secondary Focus	Limiter	Limiter	Event Demand / Limiter

Build Targets

CLASSIC LEVELS	TRAINING TARGET	COGGAN iLEVELS
Recovery 1	Aerobic endurance (extensive)	1 Recovery
Endurance 2		2 Endurance
Tempo 3		3 Tempo
		4a Sweet spot
Lactate threshold (FTP) 4	Aerobic power (intensive)	4 Threshold (FTP)
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Anaerobic capacity 6	Anaerobic endurance (extensive)	6 FRC
	Anaerobic power (intensive)	7a Pmax/FRC
	Maximal power	7 Pmax
(Neuromuscular power 7)		

iLevels or Optimized Intervals are a driver

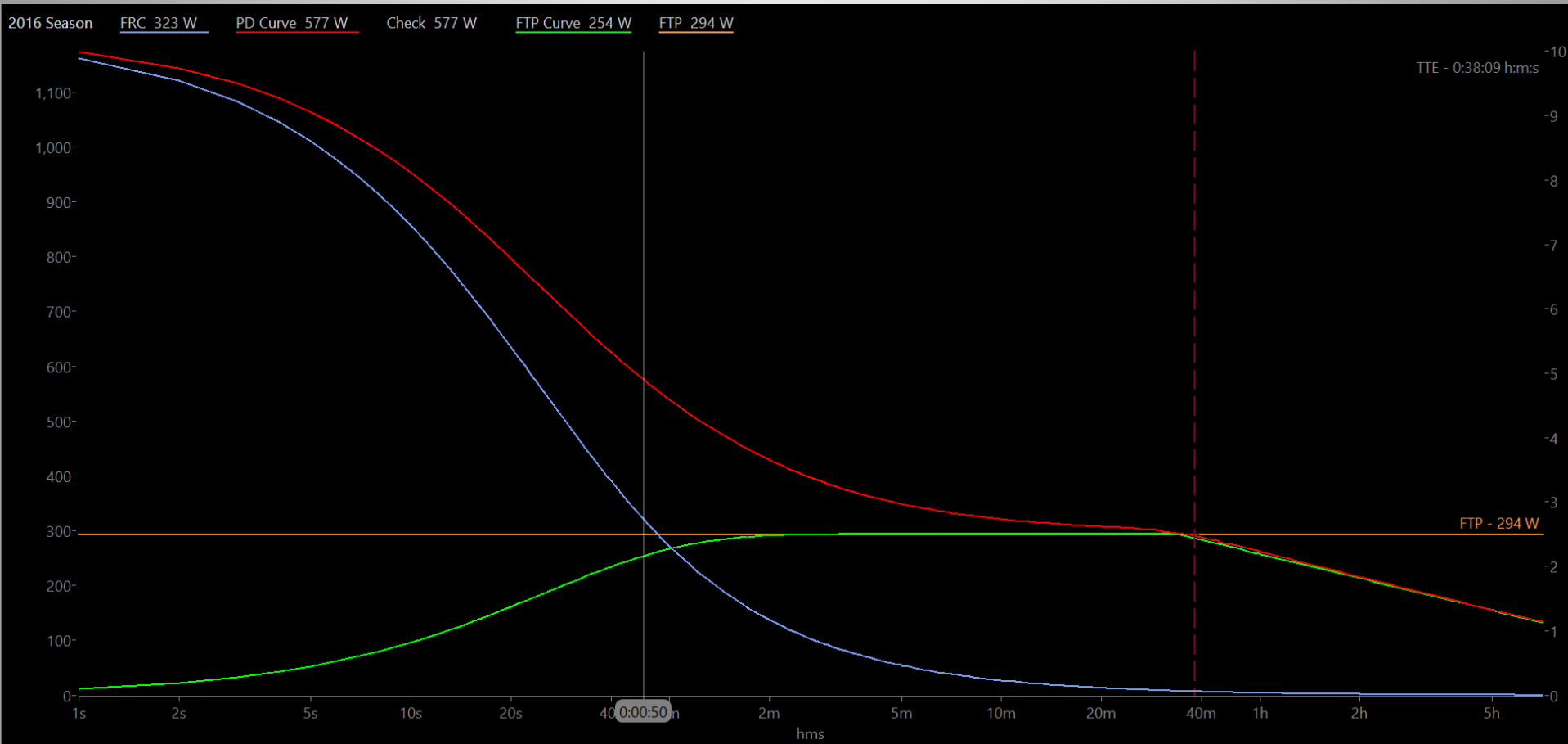
Level	Energy System	Target Duration h:m:s	Target Power W	Prescription Range W	Reps Estimate Progressive	Total TiZ Estimate Progressive	Recovery Time Estimate Ratio
Pmax	Max	0:00:10	963	+/- 15	5-10	<1 min	7-12 Min (Full)
Pmax/FRC	Intensive Anaerobic (FRC)	0:00:29	708	+/- 15	4-10	<2 mins	7-12 Min (Full)
FRC	Extensive Anaerobic (FRC)	0:00:48	588	+/- 15	4-8	4-12 mins	1:10 to 1:20 (Ratio Range)
FRC/FTP	Max Aerobic	0:04:46	353	+/- 10	3-8	12-30 mins	1:1 to 1:2 (Ratio Range)
FRC/FTP	Intensive Aerobic (FTP)	0:24:05	300	+/- 10	1-4	30-90 mins	<5 min
FTP	Extensive Aerobic (FTP)	0:38:09	294	+/- 10	1	TTE	NA



Track FRC: Battery size matters

Great Charts in the Facebook
Group right now

FRC Development



Max Aerobic & FRC



Build: Anaerobic Fatigue Resistance Workout

OPTIMIZED EXTENSIVE FRC

WU:

10-15 Min Warm Up, progressing to Zone 2

TARGET: (PW: iLevel Z2)

CADENCE: Self Selected with 2 x 1 minute Fast Pedals to wake up legs

MS1:

6 x Extensive FRC (Optimized Interval Chart)

TARGET: FRC - Duration and Intensity

REST: 7 Min between intervals

CADENCE: Self Selected, standing start for 5-10 seconds to get up to power, completed seated

Ride all other time in Zone 2

CD:

Cool Down 5-15 minutes

TARGET: (PW: iLevel Z1)

Cadence: small chain ring

OPTIMIZED INTENSIVE FRC

WU:

10-15 Min Warm Up, progressing to Zone 2

TARGET: (PW: iLevel Z2)

CADENCE: Self Selected with 2 x 1 minute Fast Pedals to wake up legs

MS1:

7 x Intensive PMAX / FRC (Optimized Interval Chart)

TARGET: PMAX/FRC - Duration and Intensity

REST: 7 Min between intervals

CADENCE: Self Selected, standing start for 5-10 seconds to get up to power, completed seated

Ride all other time in Zone 2

CD:

Cool Down 5-15 minutes

TARGET: (PW: iLevel Z1)

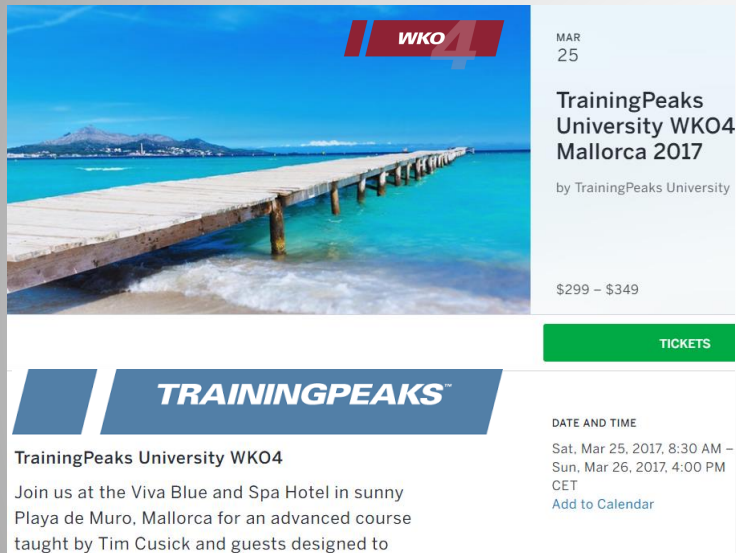
Cadence: small chain ring

Progress number of intervals

Q&A

Advanced Education

TrainingPeaks University



WKO4

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Greenville, SC with Bobby Julich in May

Chronic Fatigue

Not overtraining

Chronic Fatigue: Substrate Depletion

	Effect	Recovery
One Sprint	ATP Levels drop	Fast recovery
Several Sprints	Creatine Phosphate (CP) Levels drop	Several Minutes
Multi Sets of Sprints	Glycogen Levels drop, used to replenish ATP/CP	Hours – Days Can take several days based off diet

The interesting point is that if you train hard enough (with enough volume, intensity, and frequency) to get good adaptations, you are unlikely to replete 100% of all your glycogen each week, *particularly* if you're on a lower carb or hypocaloric diet. Oftentimes, this means that week after week of hard training leads to lower and lower glycogen levels, which have been repeatedly shown to be linked with decreases in performance

Chronic Fatigue:

CNS/Central/Neuroendocrine alterations

	Acute	Chronic
Cortisol/Testosterone	T increases	Cortisol increases, T declines
Sympathetic/Parasympathetic	Both increase	Sympathetic increases (Fight or Flight), Parasympathetic decreases (recovery and regeneration)

Chronic Fatigue: Microtrauma

	Acute	Chronic
Microtrauma	Increase but heals	Accumulates

Causes of Fatigue

- Cardiovascular / anaerobic
- Energy Supply / Depletion
- Neuromuscular Fatigue
- Muscle trauma
- Biomechanical
- Thermoregulatory
- Psychological / motivational